



BACHELOR IN SOFTWARE ENGINEERING (HONOURS)

FINAL EXAMINATION MAY 2023

Course : EC3357 (Machine Learning)

Time: 1:30 pm – 4.30 pm
(3 hours)

Lecturer : Rajesvary Rajoo

Date: 11 September 2023

Instructions:

Answer **ALL** questions in the Answer Booklet provided.

The maximum number of marks is 100.

This question paper consists of **3** printed pages.
(excluding front cover)

DO NOT OPEN THIS BOOKLET UNTIL YOU ARE TOLD TO DO

1. a. Explain any **TWO (2)** reasons for using Machine Learning **(4 MARKS)**
 - b. i. Describe what a Machine Learning is by using the terms Task (T), Performance Metric (P) and Experience (E). **(3 MARKS)**
 - ii. Illustrate the elements of T, P and E in the context of developing an image recognition system for autonomous vehicles. **(6 MARKS)**
 - c. Distinguish between regression and classification by giving an example of each. **(6 MARKS)**
 - d. Outline an example problem in which unsupervised learning is preferred over supervised learning. Justify your answer. **(6 MARKS)**

2. a. Differentiate simple linear regression and multiple linear regression. Write and explain the general equation of each. **(6 MARKS)**
 - b. What does it mean by regression line? **(3 MARKS)**
 - c. Explain the **TWO (2)** circumstances in which it is acceptable to remove outliers in X variables. **(4 MARKS)**
 - d. Describe by using an example when to use logistic regression over linear regression. **(6 MARKS)**
 - e. Explain what a decision boundary in logistic regression is. Illustrate your answer by relating the example that you have provided in 2(d). **(6 MARKS)**

3. a. Assuming that you are training a Support Vector Machine on a small dataset with 4 points shown in Figure 1. This dataset consists of two examples with class label -1 (denoted with plus), and two examples with class label +1 (denoted with triangles).

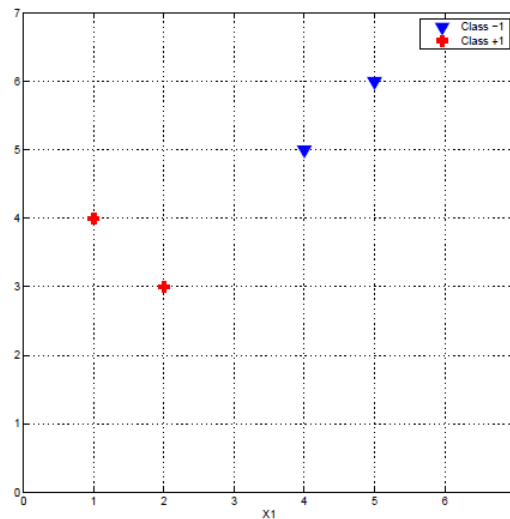


Figure 1

- i. Determine the weight vector w and bias b . Write the equation corresponding to the decision boundary. **(8 MARKS)**
 - ii. Circle the support vectors and draw the decision boundary. **(3 MARKS)**
- b. Explain the terms training data, test data and validation data **(6 MARKS)**
- c. Assuming that you are given a data set consists of 191 samples from 4 different classes as shown in the Table 1:

Class	No of samples
1	60
2	34
3	43
4	54

Table 1

The confusion matrix obtained by training a classifier and evaluating the trained model on this test set is shown in Table 2 below. Let that matrix be called " M ," and each element in the matrix be denoted by " M_{ij} ," where " i " is the row number (predicted class), and " j " is the column number (expected class), e.g., $M_{11}=52$, $M_{42}=1$.

	1	2	3	4
1	52	3	7	2
2	2	28	2	0
3	5	2	25	12
4	1	1	9	40

Table 2

Answer the following questions regarding the model's performance using the given confusion matrix.

- i. Calculate the overall accuracy of the model. **(2 MARKS)**
 - ii. Table 2 shows that $M_{24}=0$. Explain what this implies? **(3 MARKS)**
 - iii. Identify the class we should focus on to improve the model's performance. Justify your answer. **(3 MARKS)**
- 4 a. i. What is the significance of K in k-Means algorithm? **(4 MARKS)**
- ii. Justify why does k-Means Clustering mostly use the Euclidean Distance metric. **(4 MARKS)**
- iii. Identify and explain any **ONE (1)** application of k-Means Clustering. **(4 MARKS)**
- b. Differentiate Feature Selection and Feature Extraction. **(4 MARKS)**

- c. What is overfitting? What are potential causes? **(5 MARKS)**
- d. Suppose you have been given an unlabeled data set containing 1000 samples in 20 dimensions. You ran the K-means algorithms on it using 900 centroids, and you achieved a satisfying matching. However, your supervisor examines your work and states that your algorithm is overfitting the data. Explain how this is related to overfitting. **(4 MARKS)**

-END OF QUESTION PAPER-